

$$\begin{vmatrix} 0-0 & -\frac{b}{a}+\frac{\beta}{\alpha} & -\frac{d}{c}+\frac{\delta}{\gamma} \\ 1 & \frac{1}{a} & \frac{1}{c} \\ 1 & \frac{1}{\alpha} & \frac{1}{\gamma} \end{vmatrix} = 0 \quad \text{(subtract 3rd row from 2nd and first)}$$

or

$$\begin{vmatrix} 0 & (a\beta - b\alpha)/a\alpha & (c\delta - d\gamma)/c\gamma \\ 0 & (\alpha - a)/a\alpha & (\gamma - c)/c\gamma \\ 1 & 1/\alpha & 1/\gamma \end{vmatrix} = 0$$

or

$$\frac{(a\beta - b\alpha)}{a\alpha} \cdot \frac{(\gamma - c)}{c\gamma} - \frac{(c\delta - d\gamma)}{c\gamma} \cdot \frac{(\alpha - a)}{a\alpha} = 0$$

or

$$(\gamma - c)(a\beta - b\alpha) - (\alpha - a)(c\delta - d\gamma) = 0.$$

Ex. 11. Prove that the lines

$$3x - 5 = 4y - 9 = 3z \quad \text{and} \quad x - 1 = 2y - 4 = 3z$$

meet in a point and equation of the plane in which they lie is

$$3x - 8y + 3z + 13 = 0.$$

[Kashmir 75]

Ex. 12. Show that the lines

$$\frac{x+4}{3} = \frac{y+6}{5} = \frac{z-1}{-2}$$

and $3x - 2y + z + 5 = 0 = 2x + 3y + 4z - 4$

are coplanar. Find also the co-ordinates of their point of intersection and the equation of the plane in which they lie.

Equation of any plane through the line

$$3x - 2y + z + 5 = 0, \quad 2x + 3y + 4z - 4 = 0$$

is $(3x - 2y + z + 5) + \lambda(2x + 3y + 4z - 4) = 0$

or $(3 + 2\lambda)x + (-2 + 3\lambda)y + (1 + 4\lambda)z + (5 - 4\lambda) = 0.$

It is parallel to $\frac{x+4}{3} = \frac{y+6}{5} = \frac{z-1}{-2}$, then

$$3(3 + 2\lambda) + 5(-2 + 3\lambda) - 2(1 + 4\lambda) = 0.$$

$\therefore 13\lambda = 3$; or $\lambda = 3/13$. Hence the plane is

$$45x + 17y + 25z + 53 = 0.$$

The point $(-4, -6, 1)$ on the line clearly satisfies

$$\therefore 45(-4) - 17(-6) + 25(1) + 53 = 0$$

and hence the other line also lies in the plane, so that the two lines are coplanar.

*

Point of Intersection.

Any point on the line is $(3r-4, 5r-6, -2r+1)$.

As the two lines meet, this point should satisfy both the equations of the other line.

$$\therefore 3(3r-4) - 2(5r-6) + (-2r+1) + 5 = 0.$$

$$\text{or } -3r + 6 = 0; \quad \therefore r = 2$$

$$\text{and } 2(3r-4) + 3(5r-6) + 4(-2r+1) - 4 = 0$$

$$\text{or } 13r - 26 = 0; \quad \therefore r = 2.$$

Since the two values of r are equal, therefore the two lines intersect and putting $r=2$, the point of intersection is $(2, 4, -3)$.

(b) Show that the lines

$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $4x-3y+1=0=5x-3z+2$ are coplanar. Also find their point of intersection and the equation of the plane in which these lie. [Burdwan 78, Gorakhpur 79]

Ans. $\lambda = -\frac{1}{2}, x-2y+z=0; (-1, -1, -1)$.

(c) Prove that the lines

$$\frac{x-9}{2} = \frac{y+4}{-1} = z-5 \text{ and}$$

$6x+4y-5z=4, x-5y+2z=12$ are coplanar. Find the equation of the plane in which these lie and show that the point of intersection is $(3, -1, 2)$. [Agra 80, Kanpur 80]

Ans. $\lambda = 1/2, x+y-z=0, (3, -1, 2)$

Ex. 14. Prove that the lines

$$x+2y-5z+9=0=3x-y-2z-5$$

$$\text{and } 2x+3y-z-3=0=4x-5y+z+3$$

are coplanar.

[Berhampur 81]

The lines will be coplanar if

$$\Delta = \begin{vmatrix} 1 & 2 & -5 & 9 \\ 3 & -1 & -2 & -5 \\ 2 & 3 & -1 & -3 \\ 4 & -5 & 1 & 3 \end{vmatrix} = 0 \quad [\S 6 (c) P. 91]$$

Applying R_2-3R_1, R_3-2R_1 and R_4-4R_1 .

$$\Delta = \begin{vmatrix} 1 & 2 & -5 & 9 \\ 0 & -7 & 17 & -32 \\ 0 & -1 & 9 & -21 \\ 0 & -13 & 21 & -33 \end{vmatrix} = 0 \text{ or } \begin{vmatrix} 7 & 17 & 32 \\ 1 & 9 & 21 \\ 13 & 21 & 33 \end{vmatrix} = 0$$

$$\text{or } 7(297 - 441) - 1(561 - 672) + 13(357 - 288) = 0$$

$$\text{or } 7(-144) - 1(-111) + 13(69) = 0.$$

$$\text{or } -1008 + 111 + 897 = 0 \text{ or } -1008 + 1008 = 0.$$

Hence the lines are coplanar.

Ex. 15. Find the equation of the plane through the line

$$\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$$

and perpendicular to the plane containing

$$\frac{x}{m} = \frac{y}{n} = \frac{z}{l} \text{ and } \frac{x}{n} = \frac{y}{l} = \frac{z}{m}.$$

[Jiwaji 67; Allahabad 67; Ranchi 79; Luck. 82; Agra 75;
Gorakhpur 73; Pb. 73]

Equation of the plane through the first line is

$$\text{where } \begin{cases} Ax + By + Cz = 0 \\ Al + Bm + Cn = 0 \end{cases} \quad \left\{ \begin{array}{l} [\S 5 \text{ P. } 88] \\ \dots(1) \\ \dots(2) \end{array} \right.$$

Again the equation of the plane through the other two lines is

$$\begin{vmatrix} x & y & z \\ m & n & l \\ n & l & m \end{vmatrix} = 0 \quad [\S 5]$$

$$\text{or } (mn - l^2)x + (ln - m^2)y + (lm - n^2)z = 0. \quad \dots(3)$$

Now planes (1) and (3) are perpendicular, and hence

$$A(mn - l^2) + B(ln - m^2) + C(lm - n^2) = 0. \quad \dots(4)$$

Solving (2) and (3), we have

$$\frac{A}{m(lm - n^2) - n(ln - m^2)} = \frac{B}{n(lm - n^2) - m(ln - m^2)} = \frac{C}{l(mn - l^2) - m(ln - m^2)}$$

Now the denominator of A is $l(m^2 - n^2) + mn(m - n)$

$$\text{or } (m - n)(lm + mn + nl).$$

$$\therefore \frac{A}{(m - n) \Sigma lm} = \frac{B}{(n - l) \Sigma lm} = \frac{C}{(l - m) \Sigma lm}$$

or
$$\frac{A}{m-n} = \frac{B}{n-l} = \frac{C}{l-m}.$$

Hence from (1), the equation of the required plane is

$$(m-n)x + (n-l)y + (l-m)z = 0.$$

Ex. 16. Find the foot and hence the length of the perpendicular from the point (5, 7, 3) to the line

$$\frac{x-15}{3} = \frac{y-29}{8} = \frac{5-z}{5}.$$

Find the equation of the plane in which the perpendicular and the given straight line lie. [Raj. 58]

Let P be the point (5, 7, 3) and foot of perpendicular N from it to the line $\frac{x-15}{3} = \frac{y-29}{8} = \frac{5-z}{5} = r$ (say) be

$$(3r+15, 8r+29, 5-5r).$$

D.R.'s of PN are $(3r+10, 8r+22, 2-5r)$ (2)

$$\therefore 3(3r+10) + 8(8r+22) + 5(2-5r) = 0; \therefore r = -\frac{9}{2}.$$

Putting for r in (1) and (2), the point N is $(\frac{3}{2}, -7, \frac{55}{2})$ and D.R.'s of PN are $-\frac{7}{2}, -14, \frac{49}{2}$ or 1, 4, -7.

$$\therefore \text{Equation of } PN \text{ are } \frac{x-5}{1} = \frac{y-7}{4} = \frac{z-3}{-7}.$$

Also PN is distance between

$$(5, 7, 3) \text{ and } (\frac{3}{2}, -7, \frac{55}{2}) = \frac{7}{2}\sqrt{(66)}.$$

Again the plane containing given line and PN is

$$\begin{vmatrix} x-15 & y-29 & 5-z \\ 3 & 8 & 5 \\ 1 & 4 & -7 \end{vmatrix} = 0 \quad (\S 6 \text{ P. } 104)$$

Ex. 17. $A, A'; B, B'; C, C'$ are points on the axes. Show that the lines of intersection of the planes $A'BC, AB'C'; B'CA, BC'A'; C'AB, CA'B'$ are coplanar. [Banaras 41; Agra 46]

The equations of the planes $A'BC$ and $AB'C'$ (intercepts form) are respectively

$$\frac{x}{a'} + \frac{y}{b} + \frac{z}{c} = 1 \quad \text{and} \quad \frac{x}{a} + \frac{y}{b'} + \frac{z}{c'} = 1.$$

The two equations together represent the equation of the line of intersection of the two planes which will lie in the plane

$$\left(\frac{x}{a'} + \frac{y}{b} + \frac{z}{c} - 1\right) + \lambda \left(\frac{x}{a} + \frac{y}{b'} + \frac{z}{c'} - 1\right) = 0$$

for some value of λ . Let us choose $\lambda=1$ and we find that the line of intersection lies in the plane

$$x \left(\frac{1}{a} + \frac{1}{a'} \right) + y \left(\frac{1}{b} + \frac{1}{b'} \right) + z \left(\frac{1}{c} + \frac{1}{c'} \right) = 2.$$

The symmetry of the result shows that the lines of intersection of the other two pairs of planes also lie in the above plane and hence these pairs of lines are coplanar.

§ 7. To find the equation of a straight line intersecting two given lines.

Case I. Given lines in general form.

Let the equations of the line be

$$u_1 = 0, v_1 = 0 \text{ and } u_2 = 0, v_2 = 0;$$

then the equation of the required line intersecting both the above lines is

$$u_1 + \lambda_1 v_1 = 0 \text{ and } u_2 + \lambda_2 v_2 = 0.$$

The values of λ_1 and λ_2 are found by an additional given condition.

Case II. Given lines in symmetrical form.

Let the equations of the given lines be

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1} = r_1 \text{ say}$$

and $\frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2} = r_2 \text{ say.}$

Any point on (1) is $(l_1 r_1 + x_1, m_1 r_1 + y_1, n_1 r_1 + z_1)$.

Any point on (2) is $(l_2 r_2 + x_2, m_2 r_2 + y_2, n_2 r_2 + z_2)$.

The required line which will intersect the given lines will be the line joining these two points. The values of r_1 and r_2 will be found by additional given conditions.

2nd Method. Determine two planes through the given point and each containing one of the given lines. These two planes together will represent the required equations to the lines.

Exercise 12

Ex 1. Find the equations to the straight line drawn from the origin to intersect the lines

$$3x + 2y + 4z - 5 = 0 = 2x - 3y + 4z + 1$$

and $2x - 4y + z + 6 = 0 = 3x - 4y + z - 3.$

[Pb. 42]

The equations of the required lines are given by

$$(3x + 2y + 4z - 5) + \lambda_1 (2x - 3y + 4z + 1) = 0$$

and $(2x - 4y + z + 6) + \lambda_2 (3x - 4y + z - 3) = 0.$

If the above line passes through origin, then

$$-5 + \lambda_1 \cdot 1 = 0 \quad \therefore \lambda_1 = 5; \text{ and } 6 + \lambda_2 (-3) = 0, \quad \therefore \lambda_2 = 2.$$

Putting the values of λ_1 and λ_2 and simplifying, the equations of the required line are

$$13x - 13y + 24z = 0 = 8x - 12y + 13z \text{ or } \frac{x}{249} = \frac{y}{153} = \frac{z}{-52}.$$

Ex. 2. Find the equations to the planes through the point $(1, 0, -1)$ and the lines $4x - y - 13 = 0 = 3y - 4z - 1$ and $y - 2z + 2 = 0 = x - 5$ and show that the equations to the lines through the given point which intersects the two given lines can be written as $x = y + 1 = z + 2$.

Proceeding as in Ex. 1.

The required equations of the plane are found to be

$$x + 2y - 3z - 4 = 0 \text{ and } x + y - 2z - 3 = 0. \quad \dots(3)$$

These equations together represent the equations of the line. Both of them are satisfied by $(0, -1, -2)$ and the direction-ratios of the lines represented by them can be shown to be $(1, 1, 1)$. Hence it can be written as given.

Ex. 3. Find the equations to the line drawn parallel to

$$\frac{x}{4} = \frac{y}{1} = \frac{z}{1}$$

so as to meet the lines $z = 5x - 6 = 4y + 3$ and $z = 2x - 4 = 3y + 5$.

Equations of the line intersecting the given lines are

$$\begin{cases} (z - 5x + 6) + \lambda_1 (5x - 4y - 9) = 0 \\ (z - 2x + 4) + \lambda_2 (2x - 3y - 9) = 0 \end{cases}$$

$$\begin{aligned} \text{or} \quad & (5\lambda_1 - 5)x - 4\lambda_1 y + z + (6 - 9\lambda_1) = 0 \\ \text{and} \quad & (2\lambda_2 - 2)x - 3\lambda_2 y + z + (4 - 9\lambda_2) = 0 \end{aligned}$$

If the line given by planes (A) is parallel to $\frac{x}{4} = \frac{y}{1} = \frac{z}{1}$, then this is parallel to each of the planes of (A), i.e. perpendicular to their normals

$$\therefore (5\lambda_1 - 5) \cdot 4 - 4\lambda_1 \cdot 1 + 1 \cdot 1 = 0, \quad \therefore \lambda_1 = \frac{19}{8}$$

$$\text{and} \quad (2\lambda_2 - 2) \cdot 4 - 3\lambda_2 \cdot 1 + 1 \cdot 1 = 0, \quad \therefore \lambda_2 = \frac{7}{5}.$$

Putting for λ_1 and λ_2 the required equations of the line are

$$15x - 76y + 16z - 75 = 0, \quad 4x - 21y + 5z - 43 = 0.$$

Ex. 4. A line with direction cosines proportional to 2, 1, 2 meets each of the lines given by the equations.

$$x = y + a = z, \quad x + a = 2y = 2z.$$

Find the co-ordinates of each of the points of intersection.

The given lines can be written as

$$\frac{x}{1} = \frac{y+a}{1} = \frac{z}{1} = r_1 \text{ and } \frac{x+a}{2} = \frac{y}{1} = \frac{z}{1} = r_2.$$

Any point on (1) is $r_1, r_1 - a, r_1$ say P_1

Any point on (2) is $2r_2 - a, r_2, r_2$ say P_2 .

D. R.'s of P_1P_2 are $r_1 - 2r_2 + a, r_1 - a - r_2, r_1 - r_2$.

But its direction-ratios are given to be 2, 1, 2.

$$\therefore \frac{r_1 - 2r_2 + a}{2} = \frac{r_1 - a - r_2}{1} = \frac{r_1 - r_2}{2}.$$

$$\therefore (r_1 - 2r_2 + a) = 2(r_1 - a - r_2)$$

$$\text{and } 2(r_1 - a - r_2) = 1(r_1 - r_2)$$

$$\text{or } r_1 = 3a \text{ and } r_1 - r_2 = 2a; \therefore r_2 = a.$$

$$\therefore P_1 \text{ is } (3a, 2a, 3a) \text{ and } P_2 \text{ is } (a, a, a).$$

$$\text{Also } P_1P_2 \text{ is } \frac{x-a}{2} = \frac{y-a}{1} = \frac{z-a}{2}.$$

Ex. 5. Find the equations of the line intersecting the lines $\frac{x-a}{1} = \frac{y}{1} = \frac{z-a}{1}$, $\frac{x+a}{1} = \frac{y}{1} = \frac{z+a}{2}$ and parallel to the line

$$\frac{x-a}{2} = \frac{y-a}{1} = \frac{z-2a}{3}.$$

The required line being parallel to given line will have its D.R.'s 2, 1, 3. Now proceed as above.

$$P_1 \text{ is } (a, 0, a) \text{ and } P_2 \text{ is } (-3a, -2a, -5a)$$

$$\text{and } P_1P_2 \text{ is } \frac{x-a}{2} = \frac{y}{1} = \frac{z-a}{3}.$$

Ex. 6. A line with direction-cosines proportional to (2, 7, -5) is drawn to intersect the lines

$$\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1} \text{ and } \frac{x+3}{-3} = \frac{y-3}{2} = \frac{z-6}{4}.$$

Find the co-ordinates of points of intersection and the length intercepted on it.

Kanpur 80; Luck 79 B. Khand 78]

[Ans. (2, 8, -3), (0, 1, 2), $\sqrt{78}$)]

Ex. 7. Find the equation of the line which can be drawn from the point (2, -1, 3) to intersect the lines

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}, \frac{x-4}{4} = \frac{y}{5} = \frac{z+3}{3}.$$

[Rajputana 67; 57]

Equation of the plane containing the 1st line is

$$A(x-1) + B(y-2) + C(z-3) = 0, \quad \dots(1)$$

where $2A + 3B + 4C = 0$.
Also it passes through $(2, -1, 3)$.
 $\therefore A - 3B + 0C = 0$.

Solving (2) and (3), we get

$$\frac{A}{12} = \frac{B}{4} = \frac{C}{-9}.$$

Hence the plane is $12x + 4y - 9z + 7 = 0$ (4)

Similarly the plane through the 2nd line and the point $(2, -1, 3)$ is $11x - 10y + 2z - 38 = 0$

Planes (4) and (5) together give the equations to the required line.

Ex. 8. Find the equations to the straight line drawn through the origin which will intersect both the lines

$$\frac{x-1}{1} = \frac{y+3}{4} = \frac{z-5}{3}; \frac{x-4}{2} = \frac{y+3}{2} = \frac{z-14}{4}$$

Ans. $29x - 7y - 10z = 0$, $9x - 2y - 3z = 0$ or $\frac{x}{1} = \frac{y}{-3} = \frac{z}{5}$

(b) Find the equations of the line through the point $(-4, 3, 1)$ parallel to the plane $x + 2y - z = 5$ so as to intersect the line $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z-2}{-1}$. Find also the point of intersection.

Let P be $(-4, 3, 1)$ and Q be the point on the 2nd line whose coordinates are $-3r-1, 2r+3, -r+2$

Q is the point where the line through P parallel to plane $x + 2y - z = 5$ meets it.

$D.R.$'s of PQ are $3r-3, -2r, r-1$ (I)

Since it is parallel to plane and hence perpendicular to normal $1, 2, -1$

$$\therefore 1 \cdot (3r-3) + 2(-2r) - 1(r-1) = 0$$

or $-2r-2=0 \therefore r=-1$ Hence Q is $(2, 1, 3)$

and from (I) $D.R.$'s of PQ are $-6, 2, -2$ or $3, -1, 1$

$$\therefore \text{Equations to } PQ \text{ are } \frac{x+4}{3} = \frac{y-3}{-1} = \frac{z-1}{1}$$

Ex. 9. Find the equations of the straight line through the origin and cutting each of the lines

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1} \text{ and } \frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2}$$

[Punjab Hons. 41]

The equation of a plane containing first line is

$$A(x-x_1)+B(y-y_1)+C(z-z_1)=0 \quad \dots(1)$$

where

$$Al_1+Bm_1+Cn_1=0$$

It passes through $(0, 0, 0)$

$$\therefore Ax_1+By_1+Cz_1=0 \quad \dots(3)$$

Solving (2) and (3), we get

$$\frac{A}{m_1z_1-n_1y_1} = \frac{B}{n_1x_1-l_1z_1} = \frac{C}{l_1y_1-m_1x_1} \quad \dots(4)$$

$\therefore$ required plane is $Ax+By+Cz=Ax_1+By_1+Cz_1$
or $(m_1z_1-n_1y_1)x+(n_1x_1-l_1z_1)y+(l_1y_1-m_1x_1)z=0$
by (3) and (4).

Similarly the plane through origin and 2nd line is

$$(m_2z_2-n_2y_2)x+(n_2x_2-l_2z_2)y+(l_2y_2-m_2x_2)z=0.$$

These two planes together represent the required line.

Ex. 10. Show that the equation of the line through (a, b, c) which is parallel to the plane $lx+my+nz=0$ and intersects the line

$$A_1x+B_1y+C_1z+D_1=0=A_2x+B_2y+C_2z+D_2$$

are

$$l(x-a)+m(y-b)+n(z-c)=0$$

and

$$\frac{A_1x+B_1y+C_1z+D_1}{A_1a+B_1b+C_1c+D_1} = \frac{A_2x+B_2y+C_2z+D_2}{A_2a+B_2b+C_2c+D_2}$$

[Gauhati Hons. 63]

The required line is the line of intersection of the following two planes:

(1) The plane through (a, b, c) and parallel to plane $lx+my+nz=0$ and its equation therefore is

$$l(x-a)+m(y-b)+n(z-c)=0. \quad \dots(1)$$

(2) The plane through (a, b, c) and the line

$$A_1x+B_1y+C_1z+D_1=0=A_2x+B_2y+C_2z+D_2=0$$

whose equation will be found by using $P+\lambda Q=0$ and passing through (a, b, c) we find λ . Its equation will be as given. This plane together with plane (1) gives the equations of the required line.

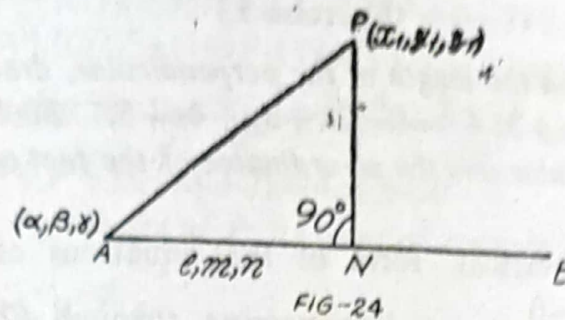
§ 8. Perpendicular distance of a point from a line and foot of the perpendicular.

[Utkal 69 ; Rewa 71, 74 ; Vikram 71 Marthwada 71]

Let the equation of the line in symmetrical form be

$$\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$$

and the point be $P(x_1, y_1, z_1)$ from where the perpendicular PN is drawn. We have to find the length PN i.e. p .



Now from the right-angled $\triangle APN$, we get

$$p^2 = AP^2 - AN^2, \quad \dots(1)$$

$$AP^2 = (x_1 - \alpha)^2 + (y_1 - \beta)^2 + (z_1 - \gamma)^2$$

AN is the projection of AP joining (α, β, γ) and (x_1, y_1, z_1) on AP whose actual direction-cosines are l, m, n and hence by § 8 P 21.

$$AN = (x_1 - \alpha)l + (y_1 - \beta)m + (z_1 - \gamma)n,$$

$$\therefore p^2 = [(x_1 - \alpha)^2 + (y_1 - \beta)^2 + (z_1 - \gamma)^2] - [(x_1 - \alpha)l + (y_1 - \beta)m + (z_1 - \gamma)n]^2$$

Another Form. Since $l^2 + m^2 + n^2 = 1$

$$p^2 = [(x_1 - \alpha)^2 + (y_1 - \beta)^2 + (z_1 - \gamma)^2] (l^2 + m^2 + n^2) - [(x_1 - \alpha)l + (y_1 - \beta)m + (z_1 - \gamma)n]^2$$

By using Lagrange's identity P. 23 we get

$$p^2 = \{l(y_1 - \beta) - m(x_1 - \alpha)\}^2 + \{m(z_1 - \gamma) - n(y_1 - \beta)\}^2 + \{n(x_1 - \alpha) - l(z_1 - \gamma)\}^2$$

$$= \begin{vmatrix} x_1 - \alpha & y_1 - \beta \\ l & m \end{vmatrix}^2 + \begin{vmatrix} y_1 - \beta & z_1 - \gamma \\ m & n \end{vmatrix}^2 + \begin{vmatrix} z_1 - \gamma & x_1 - \alpha \\ n & l \end{vmatrix}^2$$

Note. While writing the equations of the line, we should use actual direction-cosines.

Note. If however, you use direction-ratios a, b, c , then you should divide the R.H.S. of above $(a^2 + b^2 + c^2)$.

Alternative method.

The foot of the perpendicular.

If N be the foot of the perpendicular, then it being a point on AB , its co-ordinates can be written as

$$(\alpha + lr_1, \beta + mr_1, \gamma + nr_1) \text{ and } P \text{ is the point } (x_1, y_1, z_1).$$

Now find the direction ratios of PN and then apply the condition that PN is perpendicular to AB . This condition will give the value of r and the co-ordinates of the foot of the perpendicular N can be found out. Also we can find the length PN and the equation of the line PN .

Exercise 13

Ex. 1. Find the length of the perpendicular, drawn from origin to the line $x+2y+3z+4=0=2x+3y+4z+5$. Find the equations of this perpendicular and the co-ordinates of the foot of the perpendicular. [Punjab 39]

The symmetrical form of the equations of the line is $\frac{x-2}{1} = \frac{y+3}{-2} = \frac{z-0}{1}$ i.e. a line passing through $(2, -3, 0)$ and direction-ratios $1, -2, 1$. If N be the foot of perpendicular, then it being any point on the above line, its co-ordinates are $(r+2, -2r-3, r)$. If P be the point $(0, 0, 0)$, then direction-ratios of PN are $(r+2, -2r-3, r)$. Now PN is perpendicular to the above line; therefore

$$1(r+2) - 2(-2r-3) - 1 \cdot r = 0 \text{ or } 6r+8=0; \quad \therefore r = -4/3$$

Hence the coordinates of the point N are

$$\left(\frac{2}{3}, -\frac{1}{3}, -\frac{4}{3}\right) \text{ and } PN \text{ is } \left(\frac{4}{9} + \frac{1}{9} + \frac{16}{9}\right) = \frac{1}{3}\sqrt{21}$$

and the equations of the line PN are $\frac{x}{2} = \frac{y}{-1} = \frac{z}{-4}$.

Note. If we are to find simply the distance then we should follow the method below.

The given equation can be reduced to symmetrical form as $\frac{x-2}{1} = \frac{y+3}{-2} = \frac{z-0}{1}$. Here $1, -2, 1$ are D.R.'s and not D.C.'s.

If p be the perpendicular from origin, then by § 8, we have

$$p^2 = \frac{1}{(a^2+b^2+c^2)} \left[\begin{vmatrix} x_1-\alpha & y_1-\beta \\ a & b \end{vmatrix}^2 + \begin{vmatrix} y_1-\beta & z_1-\gamma \\ b & c \end{vmatrix}^2 + \begin{vmatrix} z_1-\gamma & x_1-\alpha \\ c & a \end{vmatrix}^2 \right]$$

$$\text{or } p^2 = \frac{1}{1+4+1} \left[\begin{vmatrix} 0-2 & 0+3 \\ 1 & -2 \end{vmatrix}^2 + \begin{vmatrix} 0+3 & 0 \\ -2 & 1 \end{vmatrix}^2 + \begin{vmatrix} 0 & 0-1 \\ 1 & 1 \end{vmatrix}^2 \right]$$

$$\text{or } p^2 = \frac{1}{6} (1+9+4) = \frac{7}{3} = \frac{21}{9}; \quad \therefore p = \frac{1}{3}\sqrt{21}$$

Ex. 2. (a). Find the perpendicular distance from the point $(-1, 3, 9)$ to the line $\frac{x-13}{5} = \frac{y+8}{-8} = \frac{z-31}{1}$. [Delhi Hons. 54]

$$p^2 = \frac{1}{(5^2+8^2+1^2)} \left[\begin{vmatrix} -1-13 & 3+8 \\ 5 & -8 \end{vmatrix}^2 + \begin{vmatrix} 3+8 & 9-31 \\ -8 & 1 \end{vmatrix}^2 + \begin{vmatrix} 9-31 & -1-13 \\ 1 & 5 \end{vmatrix}^2 \right]$$

$$\text{or } p^2 = (1/90) [(112-55)^2 + (11-176)^2 + (-110+14)^2]$$

$$\text{or } p^2 = (1/90) [(57)^2 + (165)^2 + (96)^2] : \therefore p = \text{etc.}$$

(b) Show that the line $\frac{x-7}{2} = \frac{y-4}{7} = \frac{z-13}{10}$ touches the sphere

$$S \equiv x^2 + y^2 + z^2 - 6x + 2y - 4z + 5 = 0$$

Find the co-ordinates of P , its point of contact with S .

[Calicut 75]

Refer Q. 16. P. 115.

Let us suppose that the line touches at P any point $(2r+7, 7r+4, 10r+13)$ on the given line. The centre of the sphere is $(3, -1, 2)$ and the radius is 3. Then CP is perpendicular to given line.

D.R.'s of CP are $2r+7-3, 7r+4+1, 10r+13-2$

i.e. $2r+4, 7r+5, 10r+11$ and that of line are 2, 7, 10.

Condition of perpendicularity gives

$$2(2r+4) + 7(7r+5) + 10(10r+11) = 0$$

$$\text{or } 153r + 153 = 0 \quad \therefore r = -1.$$

Hence the point P on putting for r is $(5, -3, 3)$.

Again $CP^2 = 4 + 4 + 1 = 9 \quad \therefore CP = 3 = \text{radius.}$

Therefore the given lines touch at the point $(5, -3, 3)$

Note. If you first want to show that the line touches then find its perpendicular distance from centre as in Ex. 2. P. 122, or § 8 P. 120, and then find the point P as above.

Ex. 3. Find the locus of a point which moves so that its distance from the line $x=y=z$ is twice its distance from the plane

$$x+y+z=1.$$

Let the given point be (α, β, γ) and the line $\frac{x}{1} = \frac{y}{1} = \frac{z}{1}$,

$$\therefore p_1^2 = \frac{1}{(1+1+1)} \left[\begin{vmatrix} \alpha & \beta \\ 1 & 1 \end{vmatrix}^2 + \begin{vmatrix} \beta & \gamma \\ 1 & 1 \end{vmatrix}^2 + \begin{vmatrix} \gamma & \alpha \\ 1 & 1 \end{vmatrix}^2 \right]$$

$$\text{or } p_1^2 = \frac{(\alpha-\beta)^2 + (\beta-\gamma)^2 + (\gamma-\alpha)^2}{3}$$

If p_2 be the distance of (α, β, γ) from $x+y+z-1=0$.

then $p_2 = \frac{(\alpha+\beta+\gamma-1)}{\sqrt{3}}$. But $p_1 = 2p_2$ given or $p_1^2 = 4p_2^2$.

$$\therefore \frac{(\alpha-\beta)^2 + (\beta-\gamma)^2 + (\gamma-\alpha)^2}{3} = 4 \frac{(\alpha+\beta+\gamma-1)^2}{3}$$

$$\text{or } 2\Sigma\alpha^2 - 2\Sigma\alpha\beta = 4(\Sigma\alpha^2 + 2\Sigma\alpha\beta + 1 - 2\Sigma\alpha)$$

$$\text{or } 2\Sigma\alpha^2 + 10\Sigma\alpha\beta - 8\Sigma\alpha + 4 = 0$$

$$\text{or } \Sigma\alpha^2 + 5\Sigma\alpha\beta - 4\Sigma\alpha + 2 = 0$$

Generalize.

$$\therefore x^2 + y^2 + z^2 + 5xy + 5yz + 5zx - 4x - 4y - 4z + 2 = 0.$$

Ex. 4. Show that the equation to a right circular cone whose vertex is at the origin and whose axis has direction cosines (l, m, n) and whose semi-vertical angle is θ , is

$$\Sigma (yn - zm)^2 = \sin^2 \theta (x^2 + y^2 + z^2).$$

If P be any point on the cone and PM be its distance from

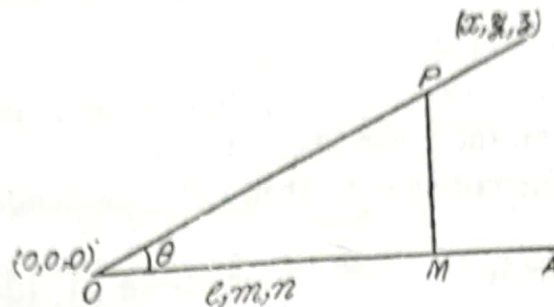


FIG-25

the axis through $(0, 0, 0)$ with direction-cosines (l, m, n) then

$$PM^2 = OP^2 \sin^2 \theta.$$

$$\text{Now } PM^2 = \left| \begin{array}{cc} x-0 & y-0 \\ l & m \end{array} \right|^2 + \left| \begin{array}{cc} y-0 & z-0 \\ m & n \end{array} \right|^2 + \left| \begin{array}{cc} z-0 & x-0 \\ n & l \end{array} \right|^2 \quad [\S 8] \quad \dots(1)$$

and $OP^2 = x^2 + y^2 + z^2.$

Hence from (1), we get the required equation as given,

Ex. 5. Find the equation to right circular cone whose vertex is $P(2, -3, 5)$ and axis the line PQ , which is equally inclined to the axes and which passes through the point $A(1, -2, 3)$.

The axis PQ is equally inclined to the axes; therefore its direction ratios are $(1, 1, 1)$.

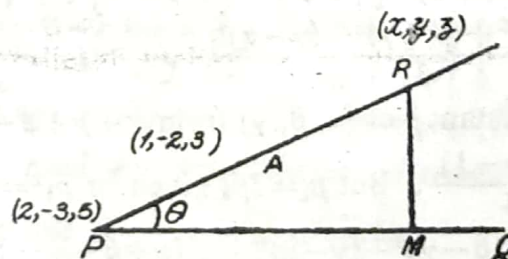


FIG-26

It passes through $(2, -3, 5)$, so that its equations are

$$\frac{x-2}{1} = \frac{y+3}{1} = \frac{z-5}{1}.$$

If $R(x, y, z)$ be any point on the cone and RM be perpendicular on the axis of the cone, then

$$RM^2 = PR^2 \sin^2 \theta \quad \dots(1)$$

where θ is the semi-vertical angle of the cone,

$$\text{Now } RM^2 = \frac{1}{3} \left\{ \begin{vmatrix} x-2 & y+3 \\ 1 & 1 \end{vmatrix}^2 + \begin{vmatrix} y+3 & z-5 \\ 1 & 1 \end{vmatrix}^2 + \begin{vmatrix} z-5 & x-2 \\ 1 & 1 \end{vmatrix}^2 \right\}$$

$$\text{or } \frac{1}{3} [(x-y-5)^2 + (y-z+8)^2 + (z-x-3)^2].$$

$$\text{Again } PR^2 = (x-2)^2 + (y+3)^2 + (z-5)^2.$$

Now direction-ratios of PA are 1, -1, 2 [§ 7. P. 19]

and direction-ratios of the axis are 1, 1, 1.

$$\therefore \cos \theta = \frac{1 \cdot 1 + (-1) \cdot 1 + 2 \cdot 1}{\sqrt{(1^2 + 1^2 + 2^2)} \cdot \sqrt{(1^2 + 1^2 + 1^2)}} = \frac{2}{\sqrt{18}};$$

$$\therefore \sin^2 \theta = 1 - \cos^2 \theta = 1 - \frac{4}{18} = \frac{14}{18} = \frac{7}{9}.$$

Hence from (1), we get the equation of the cone as

$$3 \{(y-z+8)^2 + (z-x-3)^2 + (x-y-5)^2\} = 7 [(x-2)^2 + (y+3)^2 + (z-5)^2].$$

Ex. 6. Find the equation to the right circular cone whose vertex is (2, -3, 5) axis PQ is a line equally inclined to axes and semi-vertical angle is 30° . [Calicut 82]

Exactly as above except that $\theta = 30^\circ$ given.

Ex. 7. Find the equation to the right circular cylinder of radius 2 whose axis passes through (1, 2, 3) and has direction cosines proportional to (2, -3, 6). [Garhwal 83; Madras 83; Raj. 84]

$$\text{Equations to axis are } \frac{x-1}{2} = \frac{y-2}{-3} = \frac{z-3}{6}.$$

If P be any point on the cylinder then its distance from axis is constant and equal to radius 2.

$$\therefore \frac{1}{(2^2 + 3^2 + 6^2)} \left\{ \begin{vmatrix} x-1 & y-2 \\ 2 & -3 \end{vmatrix}^2 + \begin{vmatrix} y-2 & z-3 \\ -3 & 6 \end{vmatrix}^2 + \begin{vmatrix} z-3 & x-1 \\ 6 & 2 \end{vmatrix}^2 \right\} = 2^2$$

$$(3x+2y-2)^2 + 9(2y+z-7)^2 + 4(z-3x)^2 = 196.$$

Ex. 8. The axis of a circular cylinder of radius 2 has equation

$$\frac{x-1}{2} = \frac{y}{3} = \frac{z-3}{1}, \text{ show that its equation is}$$

$$10x^2 + 5y^2 + 13z^2 - 12xy - 6yz - 4zx - 8x + 30y - 74z + 59 = 0$$

[Rohailkhand 82]

Ex. 9. Show that the equation to the right circular cone whose vertex is at the origin, whose axis has direction-cosines $\cos \alpha, \cos \beta, \cos \gamma$ and whose semi-vertical angle is θ is

$$(y \cos \gamma - z \cos \beta)^2 + (z \cos \alpha - x \cos \gamma)^2 + (x \cos \beta - y \cos \alpha)^2 = \sin^2 \theta (x^2 + y^2 + z^2). \quad [\text{Ravishanker 71}]$$

Let $P(x, y, z)$ be any point on the cone; then the direction-ratios of the line joining P to the vertex O i.e. $(0, 0, 0)$ are (x, y, z) and hence direction-cosines are

$$\frac{x}{\sqrt{x^2 + y^2 + z^2}}, \frac{y}{\sqrt{x^2 + y^2 + z^2}}, \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

D.C.'s of the axis are $\cos \alpha, \cos \beta, \cos \gamma$ and since the semi-vertical angle of cone is θ , it is the angle between the axis and OP and hence $\sin^2 \theta = \Sigma (m_1 n_2 - m_2 n_1)^2$

$$\text{or} \quad \sin^2 \theta = \frac{\Sigma (y \cos \gamma - z \cos \beta)^2}{x^2 + y^2 + z^2}$$

$$\text{or} \quad (x^2 + y^2 + z^2) \sin^2 \theta = \Sigma (y \cos \gamma - z \cos \beta)^2.$$

Ex. 10. Prove that the equation to the right circular cone whose vertex is the point $(2, -3, 5)$ and whose axis is the line which makes equal angles with the co-ordinate axes and semi-vertical angle 30° is

$$4[(y - z + 8)^2 + (z - x - 3)^2 + (x - y - 5)^2] = 3[(x - 2)^2 + (y + 3)^2 + (z - 5)^2]$$

$$\text{Here } \cos \alpha = \cos \beta = \cos \gamma = \pm \frac{1}{\sqrt{3}}.$$

[See Ex. 1 (a) P. 25].

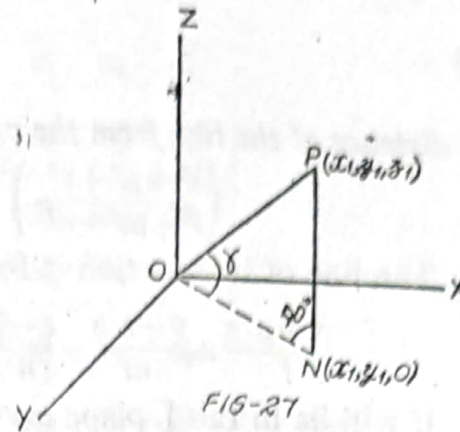
Now write down the value of $\sin^2 \theta$, where $\theta = 30^\circ$ is the angle between the axis and the line joining any point (x, y, z) to the vertex $(2, -3, 5)$ etc.

Ex. 11. A line through the origin makes angles α, β, γ with its projections on the co-ordinate planes which are rectangular. The distance of any point (x, y, z) from the line and its projections are d, a, b, c respectively, Prove that

$$d^2 = (a^2 - x^2) \cos^2 \alpha + (b^2 - y^2) \cos^2 \beta + (c^2 - z^2) \cos^2 \gamma.$$

[Burdwan Hons. 75]

Let $P(x_1, y_1, z_1)$ be any point on the given line and let N be the foot of the perpendicular from P on xy -plane; then ON is projection of OP on the xy -plane; and the angle between OP and ON is γ where N is $(x_1, y_1, 0)$



$$\therefore \cos \gamma = \frac{ON}{OP}$$

$$= \frac{\sqrt{x_1^2 + y_1^2}}{\sqrt{x_1^2 + y_1^2 + z_1^2}} \quad \dots(1)$$

Similarly,

$$\cos \alpha = \frac{\sqrt{y_1^2 + z_1^2}}{\sqrt{x_1^2 + y_1^2 + z_1^2}}, \quad \cos \beta = \frac{\sqrt{z_1^2 + x_1^2}}{\sqrt{x_1^2 + y_1^2 + z_1^2}} \quad \dots(2)$$

Lines OP and ON are

$$OP, \frac{x-0}{x_1} = \frac{y-0}{y_1} = \frac{z-0}{z_1}; \quad ON, \frac{x-0}{x_1} = \frac{y-0}{y_1} = \frac{z-0}{0}$$

Now d is the distance of a point (x, y, z) from OP .

$$\therefore d^2 = \frac{1}{(x_1^2 + y_1^2 + z_1^2)} \left[\begin{vmatrix} x & y \\ x_1 & y_1 \end{vmatrix}^2 + \begin{vmatrix} y & z \\ y_1 & z_1 \end{vmatrix}^2 + \begin{vmatrix} z & x \\ z_1 & x_1 \end{vmatrix}^2 \right]$$

$$\therefore d^2 = \frac{(xy_1 - yx_1)^2 + (yz_1 - zy_1)^2 + (zx_1 - xz_1)^2}{x_1^2 + y_1^2 + z_1^2} \quad \dots(1)$$

If c be the distance of (x, y, z) from ON i.e. $\frac{x}{x_1} = \frac{y}{y_1} = \frac{z}{0}$, then putting $z_1 = 0$ in (1), we get

$$c^2 = \frac{(xy_1 - yx_1)^2 + (y_1^2 z^2 + z^2 x_1^2)}{x_1^2 + y_1^2}$$

or
$$c^2 = z^2 + \frac{(xy_1 - yx_1)^2}{x_1^2 + y_1^2}$$

$$\therefore (c^2 - z^2) \frac{x_1^2 + y_1^2}{x_1^2 + y_1^2 + z_1^2} = \frac{(xy_1 - yx_1)^2}{x_1^2 + y_1^2 + z_1^2}$$

or
$$(c^2 - z^2) \cos^2 \gamma = \frac{(xy_1 - yx_1)^2}{(x_1^2 + y_1^2 + z_1^2)} \text{ from (1), } \dots(1)$$

Similarly,
$$(a^2 - x^2) \cos^2 \alpha = \frac{(yz_1 - zy_1)^2}{x_1^2 + y_1^2 + z_1^2} \quad \dots(5)$$

and
$$(b^2 - y^2) \cos^2 \beta = \frac{(zx_1 - xz_1)^2}{x_1^2 + y_1^2 + z_1^2}$$

Adding (4), (5), (6) and using (3), we get

$$(a^2 - x^2) \cos^2 \alpha + (b^2 - y^2) \cos^2 \beta + (c^2 - z^2) \cos^2 \gamma = d^2.$$

Ex. 12. Prove that the planes $ny - mz = \lambda$, $lz - nx = \mu$ and $mx - ly = \nu$ have a common line if $\lambda + m\mu + n\nu = 0$. Show also that

the distance of the line from the origin is

$$\left(\frac{\lambda^2 + \mu^2 + \nu^2}{l^2 + m^2 + n^2} \right)^{1/2}$$

[Agra 67]

The line of intersection is found out to be

$$\frac{x + \mu/n}{l} = \frac{y - \lambda/n}{m} = \frac{z - 0}{n}$$

If will lie in third plane $mx - ly - v = 0$ if as before

$$m(-\mu/n) - l(\lambda/n) - v = 0 \text{ or } l\lambda + m\mu + nv = 0$$

and $l(m) + m(-l) + n(0) = 0$ which is true.

In order to find the distance of the line from the origin, use § 8 P. 120.

$$\therefore d = \frac{1}{\sqrt{l^2 + m^2 + n^2}}$$

$$\left[\left| \begin{array}{cc} 0 + \mu/n & 0 - \lambda/n \\ l & m \end{array} \right|^2 + \left| \begin{array}{cc} 0 - \lambda/n & 0 \\ m & n \end{array} \right|^2 + \left| \begin{array}{cc} 0 & 0 + \mu/n \\ n & l \end{array} \right|^2 \right]^{1/2}$$

or

$$d = \frac{1}{\sqrt{l^2 + m^2 + n^2}} \left[\left(\frac{m\mu + \lambda l}{n} \right)^2 + \lambda^2 + \mu^2 \right]^{1/2}$$

Now put $m\mu + \lambda l = -uv$ $\therefore \lambda l + m\mu + nv = 0$.

$$\therefore d = \left[\frac{\lambda^2 + \mu^2 + \nu^2}{l^2 + m^2 + n^2} \right]^{1/2}$$

§ 9. Intersection of three planes.

[Agra 40]

To find the condition that the planes

$$a_1x + b_1y + c_1z + d_1 = 0, \quad \dots(1)$$

$$a_2x + b_2y + c_2z + d_2 = 0, \quad \dots(2)$$

$$a_3x + b_3y + c_3z + d_3 = 0. \quad \dots(3)$$

(a) May have a common line of intersection.

(Sagar 64 ; Agra 67 ; 68 ; Rajasthan 75, 55 ; Nagpur (63)

(b) may form a triangular prism,

(c) may intersect at a point.

Notation. If in a determinant the number of columns is not equal to the number of rows, then it is called a rectangular array. Now consider the array.

$$\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{vmatrix}$$

$\Delta_1, \Delta_2, \Delta_3, \Delta_4$ will stand for the determinants obtained after omitting the 1st, 2nd, 3rd and 4th columns respectively in the above array. Thus

$$\Delta_1 = \begin{vmatrix} b_1 & c_1 & d_1 \\ b_2 & c_2 & d_2 \\ b_3 & c_3 & d_3 \end{vmatrix}, \quad \Delta_2 = \begin{vmatrix} a_1 & c_1 & d_1 \\ a_2 & c_2 & d_2 \\ a_3 & c_3 & d_3 \end{vmatrix}$$

$$\Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}, \quad \Delta_4 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

The equation of the line of intersection of planes (1) and (2) in symmetrical form is

$$\frac{x - \frac{b_1 d_2 - b_2 d_1}{a_1 b_2 - a_2 b_1}}{\frac{b_1 c_2 - b_2 c_1}{c_1 a_2 - c_2 a_1}} = \frac{y - \frac{a_1 d_2 - a_2 d_1}{a_1 b_2 - a_2 b_1}}{\frac{a_1 c_2 - a_2 c_1}{a_1 b_2 - a_2 b_1}} = \frac{z - 0}{\frac{a_1 b_2 - a_2 b_1}{a_1 b_2 - a_2 b_1}} \quad [\S 3, P. 81] \dots (4)$$

(a) If the three planes intersect in a line, then the line of intersection of (1) and (2), i.e. (4) should lie in the plane (3), the condition for which are [$\S 4$ (c) P. 83]: (i) the line should be perpendicular to the normal of the plane; (ii) any point on the line should lie on the plane.

The first of these conditions gives

$$a_3 (b_1 c_2 - b_2 c_1) + b_3 (c_1 a_2 - c_2 a_1) + c_3 (a_1 b_2 - a_2 b_1) = 0$$

or
$$a_1 (b_2 c_3 - b_3 c_2) - b_1 (a_2 c_3 - a_3 c_2) + c_1 (a_2 b_3 - a_3 b_2) = 0,$$

i.e.
$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0, \quad \text{i.e. } \Delta_4 = 0$$

The second condition gives that the point

$$\left(\frac{b_1 d_2 - b_2 d_1}{a_1 b_2 - a_2 b_1}, \frac{a_1 d_2 - a_2 d_1}{a_1 b_2 - a_2 b_1}, 0 \right)$$

should satisfy the equation of the plane (3)

i.e.
$$a_3 x + b_3 y + c_3 z + d_3 = 0$$

$$\therefore a_3 (b_1 d_2 - b_2 d_1) + b_3 (a_1 d_2 - a_2 d_1) + d_3 (a_1 b_2 - a_2 b_1) = 0$$

i.e.
$$a_1 (b_2 d_3 - b_3 d_2) - b_1 (a_2 d_3 - a_3 d_2) + d_1 (a_2 b_3 - a_3 b_2) = 0$$

or
$$\begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix} = 0, \quad \text{i.e. } \Delta_3 = 0.$$

$$\Delta_1 = \begin{vmatrix} b_1 & c_1 & d_1 \\ b_2 & c_2 & d_2 \\ b_3 & c_3 & d_3 \end{vmatrix}, \quad \Delta_2 = \begin{vmatrix} a_1 & c_1 & d_1 \\ a_2 & c_2 & d_2 \\ a_3 & c_3 & d_3 \end{vmatrix}$$

$$\Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}, \quad \Delta_4 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

The equation of the line of intersection of planes (1) and (2) in symmetrical form is

$$\frac{x - \frac{b_1 d_2 - b_2 d_1}{a_1 b_2 - a_2 b_1}}{\frac{b_1 c_3 - b_2 c_1}{c_1 a_2 - c_2 a_1}} = \frac{y - \frac{a_2 d_1 - a_1 d_2}{a_1 b_2 - a_2 b_1}}{\frac{c_1 a_2 - c_2 a_1}{a_1 b_2 - a_2 b_1}} = \frac{z - 0}{\frac{a_1 b_2 - a_2 b_1}{a_1 b_2 - a_2 b_1}} \quad [\S 3, P. 81] \dots (4)$$

(a) If the three planes intersect in a line, then the line of intersection of (1) and (2), i.e. (4) should lie in the plane (3), the condition for which are [$\S 4$ (c) P. 83]: (i) the line should be perpendicular to the normal of the plane; (ii) any point on the line should lie on the plane.

The first of these conditions gives

$$a_3 (b_1 c_2 - b_2 c_1) + b_3 (c_1 a_2 - c_2 a_1) + c_3 (a_1 b_2 - a_2 b_1) = 0$$

or

$$a_1 (b_2 c_3 - b_3 c_2) - b_1 (a_2 c_3 - a_3 c_2) + c_1 (a_2 b_3 - a_3 b_2) = 0,$$

$$\text{i.e. } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0, \text{ i.e. } \Delta_4 = 0$$

The second condition gives that the point

$$\left(\frac{b_1 d_2 - b_2 d_1}{a_1 b_2 - a_2 b_1}, \frac{a_2 d_1 - a_1 d_2}{a_1 b_2 - a_2 b_1}, 0 \right)$$

should satisfy the equation of the plane (3)

$$\text{i.e. } a_3 x + b_3 y + c_3 z + d_3 = 0$$

$$\therefore a_3 (b_1 d_2 - b_2 d_1) + b_3 (a_2 d_1 - a_1 d_2) + d_3 (a_1 b_2 - a_2 b_1) = 0$$

$$\text{i.e. } a_1 (b_2 d_3 - b_3 d_2) - b_1 (a_2 d_3 - a_3 d_2) + d_1 (a_2 b_3 - a_3 b_2) = 0$$

$$\text{or } \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix} = 0, \text{ i.e. } \Delta_3 = 0.$$

Now while finding the symmetrical form of the equation of the line, given by planes (1) and (2), we took any point for which $z=0$. We could also take the point for which either $x=0$ or $y=0$. Had we taken the point on the line for which $x=0$, then this condition would have given that $\Delta_1=0$.

If we had taken $y=0$, then Δ_2 would be zero.

Hence the conditions for the planes to intersect in a line is that $\Delta_4=0$ and $\{\Delta_3=0 \text{ or } \Delta_2=0 \text{ or } \Delta_1=0\}$.

Note. It can be easily proved that when $\Delta_4=0$ and $\Delta_3=0$, then Δ_1 and Δ_2 are also zero.

(b) If the three planes form a triangular prism, then the line of intersection of the planes (1) and (2) is parallel to the third plane, (ii) but does not lie in it.

Now since the line (4) is parallel to the plane (3), i.e. it is perpendicular to the normal, we have as above $\Delta_4=0$.

Secondly since the line does not lie in the plane, $\Delta_3 \neq 0$ or $\Delta_1 \neq 0$ or $\Delta_2 \neq 0$.

Hence the condition for the three planes to form a triangular prism is that $\Delta_4=0$ and neither of $\Delta_1, \Delta_2, \Delta_3$ is zero.

Note. It will be sufficient to prove that $\Delta_4=0$ and that any of $\Delta_1, \Delta_2, \Delta_3$ is not zero.

(c) In case, the three planes meet at a point, then the line of intersection of the planes (1) and (2) should neither be parallel to the third plane nor lie in it. It should rather meet it at some finite point. Hence the condition is that $\Delta_4 \neq 0$ as $\Delta_4=0$ is the condition for the line to be parallel to the third plane or to lie in it.

Hence the condition for the planes to meet at a point is that $\Delta_4 \neq 0$.

Note. The point of intersection of the three planes is obtained by solving three equations of the planes for x, y, z .

Working Rule :

1. Calculate Δ_4 . If it is not zero, then the planes intersect in a point whose co-ordinates are obtained by solving the equations of the given planes for x, y, z .

2. In case $\Delta_4=0$, then calculate any of $\Delta_1, \Delta_2, \Delta_3$.

(a) In case any of $\Delta_1, \Delta_2, \Delta_3$ is also zero, then the planes intersect in a line.

(b) In case any of $\Delta_1, \Delta_2, \Delta_3$ is not zero, then the planes form a triangular prism.

Exercise 14

Ex. 1. Examine the nature of intersection of the planes

(i) $2x + y + z + 4 = 0, y - z + 4 = 0, 3x + 2y + z + 8 = 0;$

(ii) $x - 2y + z - 3 = 0, x + y - 2z - 3 = 0, x - z - 1 = 0;$

(iii) $2x + 3y - z - 2 = 0, 3x + 3y + z - 4 = 0; x - y + 2z + 5 = 0.$

(i) Rectangular array is
$$\begin{vmatrix} 2 & 1 & 1 & 4 \\ 0 & 1 & -1 & 4 \\ 3 & 2 & 1 & 8 \end{vmatrix}$$

$$\Delta_4 = \begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & -1 \\ 3 & 2 & 1 \end{vmatrix}$$

$$= 2(1+2) - 0 + 3(-1-1) = 6 - 6 = 0.$$

Since $\Delta_4 = 0$, the planes do not intersect at a point. They intersect in a line or form a triangular prism.

$$\Delta_3 = \begin{vmatrix} 2 & 1 & 4 \\ 0 & 1 & 4 \\ 3 & 2 & 8 \end{vmatrix} = 4 \begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 3 & 2 & 2 \end{vmatrix} = 0$$

Since Δ_3 is also zero, the given planes intersect in a line.

The equation of the line in symmetrical form can be obtained as usual from any two of the three planes.

(ii) Rectangular array is
$$\begin{vmatrix} 1 & -2 & 1 & -3 \\ 1 & 1 & -2 & -3 \\ 1 & 0 & -1 & -1 \end{vmatrix}$$

$$\Delta_4 = \begin{vmatrix} 1 & -2 & 1 \\ 1 & 1 & -2 \\ 1 & 0 & -1 \end{vmatrix}$$

$$= 1(-1+0) - 1(2-0) + 1(4-1) = -3 + 3 = 0.$$

$$\Delta_3 = \begin{vmatrix} 1 & -2 & -3 \\ 1 & 1 & -3 \\ 1 & 0 & -1 \end{vmatrix}$$

$$= 1(-1+0) - 1(2+0) + 1(6+3) = -3+9=6.$$

Since $\Delta_4=0$ but $\Delta_3 \neq 0$, hence the given set of planes form a triangular prism.

(iii) Rectangular array is $\begin{vmatrix} 2 & 3 & -1 & -2 \\ 3 & 3 & 1 & -4 \\ 1 & -1 & 2 & 5 \end{vmatrix}$

$$\Delta_4 = \begin{vmatrix} 2 & 3 & -1 \\ 3 & 3 & 1 \\ 1 & -1 & 2 \end{vmatrix}$$

$$= 2(6+1) - 3(6-1) + 1(3+3) = 14 - 15 + 6 = 5.$$

Since $\Delta_4 \neq 0$, the planes intersect at a point.

Eliminating z between 1st and 2nd, we get $5x+6y-6=0$.

Eliminating z between 1st and 3rd, we get $5x+5y-9=0$.

Solving these, we get, $x = \frac{24}{5}$, $y = -3$. Putting for x and y in any one we get $z = -\frac{7}{5}$. Hence the planes intersect at the point $(\frac{24}{5}, -3, -\frac{7}{5})$.

Ex. 2. Examine the nature of intersection of the following planes

(a) $2x+4y+2z=7$, $5x+y-z=9$, $x-y-z=6$.

(b) $x+y+z+6=0$, $x+2y+2z+6=0$, $x+3y+3z+6=0$.

[Raj. 76]

(c) $2x-3y-7z=0$, $3x-14y-13z=0$, $8x-31y-33z=0$.

[Meerut 77]

$$\Delta_4 = \begin{vmatrix} 2 & 4 & 2 \\ 5 & 1 & -1 \\ 1 & -1 & -1 \end{vmatrix} = 2(-1-1) - 5(-4+2) + 1(-4-2) = -4+10-6=0.$$

Since $\Delta_4=0$, the planes do not intersect at a point. Hence they either intersect in a line or form a triangular prism

$$\Delta_1 = \begin{vmatrix} 4 & 2 & -7 \\ 1 & -1 & -9 \\ -1 & -1 & -6 \end{vmatrix} = 4(6-9) - 1(-12-7) = -12 + 19 + 25 = 32 \neq 0$$

Since $\Delta_4 = 0$ but $\Delta_1 \neq 0$ therefore they form a triangular prism.

(b) Here both $\Delta_4 = 0$ and $\Delta_1 = 0$. Hence they meet in a line.

(c) In this case also $\Delta_4 = 0$ and $\Delta_1 = 0$. Hence they intersect in a line and the equations of the line in symmetrical form are

$$\frac{x}{59} = \frac{y}{-5} = \frac{z}{19}$$

Ex. 3. Examine the nature of intersection of planes

(i) $2x - y + z = 4$, $5x + 7y + 2z = 0$, $3x + 4y - 2z + 3 = 0$.

$\Delta_4 = -61$ i.e. $\neq 0$. Hence the planes intersect at a point which can be obtained on solving the equations to be $(1, -1, 1)$.

(ii) $x + 2y + 3z = 6$, $3x + 4y + 5z = 2$, $5x + 4y + 3z + 18 = 0$.

$\Delta_4 = 0$, $\Delta_3 = 0$. Hence the planes intersect in a line whose equation can be put as $\frac{x+10}{1} = \frac{y-8}{-2} = \frac{z}{1}$.

(iii) $2x - 5y + z = 3$, $x + y + 4z = 5$, $x + 3y + 6z = 1$.

$\Delta_4 = 0$, $\Delta_3 = 42$ i.e. $\neq 0$.

Hence the planes form a prism.

Ex. 4. For what values of λ do the planes

$x - y + z + 1 = 0$, $\lambda x + 3y + 2z - 3 = 0$, $3x + \lambda y + z - 2 = 0$.

(a) Intersect at a point (b) Intersect along a line (c) Form a triangular prism. [Punjab 74]

$$\Delta_4 = \begin{vmatrix} 1 & -1 & 1 \\ \lambda & 3 & 2 \\ 3 & \lambda & 1 \end{vmatrix} = c_1 + c_2, c_2 + c_3.$$

$$\Delta_4 = \begin{vmatrix} 0 & 0 & 1 \\ \lambda + 3 & 5 & 2 \\ 3 + \lambda & \lambda + 1 & 1 \end{vmatrix} = (\lambda + 3)(\lambda + 1) - 5(3 + \lambda) = \lambda^2 + 4\lambda + 3 - 15 - 5\lambda = \lambda^2 - \lambda - 12 = (\lambda - 4)(\lambda + 3) \dots (A)$$

$\Delta_4 = 0$ when $\lambda = 4, -3$.

Similarly we can evaluate

$\Delta_1 = -5(\lambda - 2)$, $\Delta_2 = 3\lambda - 16$, $\Delta_3 = (\lambda - 2)(\lambda + 3)$

(a) Intersect at a point $\Delta_4 \neq 0$, $\lambda \neq 4$, $\lambda \neq -3$,

(b) Intersect along a line $\Delta_4 = 0$ and any one of $\Delta_1, \Delta_2, \Delta_3$ is also zero. $\Delta_4 = 0$ for $\lambda = 4$, $\lambda = -3$.

For $\lambda = -3$, $\Delta_3 = 0$.

Thus for $\lambda = -3$, $\Delta_4 = 0$, and $\Delta_3 = 0$ and as such they intersect in a line.

(c) Form a triangular prism. $\Delta_4 = 0$ and none of $\Delta_1, \Delta_2, \Delta_3$ is

For $\lambda = 4$, $\Delta_4 = 0$ but Δ_1, Δ_2 or Δ_3 is not zero.

Hence $\lambda = 4$ gives the case for triangular prism.

Ex. 5. Prove that the planes $x + ay + (b+c)z + d = 0$, $x + by + (c+a)z + d = 0$, $x + cy + (a+b)z + d = 0$ have a common line of intersection.

[Allahabad 78, Kanpur 82]

Rectangular arrays is

$$\begin{vmatrix} 1 & a & b+c & d \\ 1 & b & c+a & d \\ 1 & c & a+b & d \end{vmatrix}, \therefore \Delta_4 = \begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a & a+b+c \\ 1 & b & a+b+c \\ 1 & c & a+b+c \end{vmatrix} = (a+b+c) \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix} = 0$$

$$\Delta_3 = \begin{vmatrix} 1 & a & d \\ 1 & b & d \\ 1 & c & d \end{vmatrix} = d \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix} = 0.$$

Since $\Delta_4 = 0$ and $\Delta_3 = 0$, the planes intersect in a line.

Ex. 6. A, B, C are points $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$ respectively. Find the equation of the plane through BC which bisects OA . By symmetry write down the equations of the planes through CA bisecting OB and through AB bisecting OC . Show that these planes pass through the point $\left(\frac{a}{4}, \frac{b}{4}, \frac{c}{4}\right)$.

[Madras 74]

Equation to line BC is

$$\frac{x}{0} = \frac{y-b}{b} = \frac{z}{-c}.$$

Any plane through BC is

$$Lx + M(y-b) + Nz = 0$$

$$L \cdot 0 + Mb - Nc = 0, \quad \dots(2)$$

where

Since it bisects OA therefore $(a/2, 0, 0)$ will satisfy (1)

$$L \cdot a/2 + M(-b) = 0, \quad \dots(3)$$

From (2) and (3), we get

$$\frac{L}{\frac{2}{a}} = \frac{M}{\frac{1}{b}} = \frac{N}{\frac{1}{c}}$$

Putting in (1) the required plane is

$$\frac{2}{a}x + \frac{1}{b}(y-b) + \frac{1}{c}z = 0$$

$$\frac{2}{a}x + \frac{y}{b} + \frac{z}{c} = 1.$$

or

Similarly other planes by symmetry are

$$\frac{x}{a} + \frac{2y}{b} + \frac{z}{c} = 1 \quad \text{and} \quad \frac{x}{a} + \frac{y}{b} + \frac{2z}{c} = 1.$$

Clearly they are all satisfied by the point $(\frac{a}{4}, \frac{b}{4}, \frac{c}{4})$.

Ex. 7. Prove that the planes $x = cy + bz$, $y = az + cx$, $z = bx + ay$ pass through one line if $a^2 + b^2 + c^2 + 2abc = 1$, and show that the line of intersection then is

$$\frac{x}{\sqrt{1-a^2}} = \frac{y}{\sqrt{1-b^2}} = \frac{z}{\sqrt{1-c^2}}.$$

[Ravishankar 71; Nagpur 62; B. khand 78; Agra 73; Raj. 77, 75; Pb. 77]

We will find the line of intersection of first two planes and show that it also lies in the third. If l, m, n , be the d.c.'s of the line of intersection of first two planes, then

$$l - mc - nb = 0 \quad \text{and} \quad -lc + m - na = 0.$$

$$\frac{l}{ac+b} = \frac{m}{bc+a} = \frac{n}{1-c^2}.$$

Also both these planes pass through origin and hence the line also passes through origin so that its equation is

$$\frac{x}{ac+b} = \frac{y}{bc+a} = \frac{z}{1-c^2}.$$

This line will lie in the third plane $bx + ay - z = 0$ if its normal is perpendicular to the line and also the point $(0, 0, 0)$ on it satis-

fies it. The second condition is evidently true and the first condition gives

$$(ac+b)b + (bc+a)a + (1-c^2)(-1) = 0$$

or

$$a^2 + b^2 + c^2 + 2abc = 1. \quad \dots(1)$$

Hence the three planes meet in the line (1). In order to put the equation in the given form, we see

$$\begin{aligned} ac+b &= \sqrt{[(ac+b)^2]} = \sqrt{(a^2c^2 + 2abc + b^2)} \\ &= \sqrt{(a^2c^2 + 1 - a^2 - c^2)} = \sqrt{[(1-a^2)(1-c^2)]} \end{aligned} \quad \text{by (2)}$$

$$\text{Similarly } bc+a = \sqrt{[(1-b^2)(1-c^2)]}$$

Hence the line (1) can be written as

$$\frac{x}{\sqrt{[(1-a^2)(1-c^2)]}} = \frac{y}{\sqrt{[(1-c^2)(1-b^2)]}} = \frac{z}{1-c^2}$$

$$\text{or } \frac{x}{\sqrt{(1-a^2)}} = \frac{y}{\sqrt{(1-b^2)}} = \frac{z}{\sqrt{(1-c^2)}}. \quad \text{Proved.}$$

Ex. 8. Show that the planes

$$cy-bz=l, \quad az-cx=m, \quad bx-ay=n$$

intersect in a line if $al+bm+cn=0$ and direction ratios of the line are a, b, c . [Berhampur 76; Punjab 80]

Let us find the line of intersection of first two planes

$$cy-bz=l \quad \text{and} \quad -cx+az=m.$$

$$\text{Putting } z=0 \text{ we get } x=-\frac{m}{c}, y=\frac{l}{c}.$$

Also if P, Q, R be the D.C.'s of the line then it is perpendicular to the normal of each and hence we have

$$0P+cQ-bR=0$$

$$-cP+0Q+aR=0$$

$$\therefore \frac{P}{ac} = \frac{Q}{bc} = \frac{R}{c^2} \quad \text{or} \quad \frac{P}{a} = \frac{Q}{b} = \frac{R}{c}$$

$\therefore$ D.R.'s are a, b, c

Hence the line given by first two planes is

$$\frac{x+m/c}{a} = \frac{y-l/c}{b} = \frac{z-0}{c}.$$

It will lie in the 3rd plane $bx-ay-n=0$, if $(-m/c, l/c, 0)$ lie on the plane

$$\text{i.e. } b\left(-\frac{m}{c}\right) - a\left(\frac{l}{c}\right) - n = 0 \quad \text{or} \quad al+bm+cn=0$$

And the line is perpendicular to normal $b, -a, 0$,

$$\therefore a(b) + b(-a) + c(0) = 0 \quad \text{or} \quad ab-ab=0$$

which is true. Hence the planes intersect in a line,

Ex. 9. Prove that the planes

$$ax+hy+gz=0, hx+by+fz=0, gx+fy+cz=0$$

have a common line of intersection if

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

and the direction-ratios of the line satisfy the equations

$$\frac{l^2}{\frac{\partial \Delta}{\partial a}} = \frac{m^2}{\frac{\partial \Delta}{\partial b}} = \frac{n^2}{\frac{\partial \Delta}{\partial c}}$$

If l, m, n be the d.c.'s of the common line, then from 1st and 2nd, we have $\frac{l}{hf-bg} = \frac{m}{gh-af} = \frac{n}{ab-h^2}$... (1)

Hence the equation of the line is

$$\frac{x}{hf-bg} = \frac{y}{gh-af} = \frac{z}{ab-h^2} \quad \dots (2)$$

It will lie in the third plane $gx+fy+cz=0$ if

$$g(hf-bg) + f(gh-af) + c(ab-h^2) = 0$$

$$\text{or } abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \text{ or } \Delta = 0. \quad \dots (3)$$

$$\therefore \frac{\partial \Delta}{\partial a} = bc - f^2, \frac{\partial \Delta}{\partial b} = ca - g^2, \frac{\partial \Delta}{\partial c} = ab - h^2 \quad \dots (4)$$

To put the direction-ratios in the given form, we see

$$\begin{aligned} (hf-bg)^2 &= h^2f^2 + b^2g^2 - 2bfg h \\ &= h^2f^2 + b^2g^2 - b(af^2 + bg^2 + ch^2 - abc) \text{ by (3)} \\ &= h^2f^2 - abf^2 - bch^2 + ab^2c \\ &= f^2(h^2 - ab) - bc(h^2 - ab) \\ \text{or } (hf-bg)^2 &= (ab-h^2)(bc-f^2). \quad \dots (5) \end{aligned}$$

$$\text{Similarly } (gh-af)^2 = (ab-h^2)(ca-g^2). \quad \dots (6)$$

Hence from (1), by the help of (5) and (6), we get

$$\frac{l^2}{(ab-h^2)(bc-f^2)} = \frac{m^2}{(ab-h^2)(ca-g^2)} = \frac{n^2}{(ab-h^2)^2}$$

$$\text{or } \frac{l^2}{\frac{\partial \Delta}{\partial a}} = \frac{m^2}{\frac{\partial \Delta}{\partial b}} = \frac{n^2}{\frac{\partial \Delta}{\partial c}} \quad \text{Proved.}$$

Ex. 10. Prove that the planes $x = y \sin \psi + z \sin \phi$, ... (1)

$$z = x \sin \phi + y \sin \theta \dots (2), y = z \sin \theta + x \sin \psi \quad \dots (2)$$

will intersect in the line

$$\frac{x}{\cos \theta} = \frac{y}{\cos \phi} = \frac{z}{\cos \psi}, \text{ if } \theta + \phi + \psi = \frac{\pi}{2}$$

[Allahabad 82; Garhwal 78]

Let us find the line of intersection of the planes (1) and (2) i.e. of $x - y \sin \psi - z \sin \phi = 0$ and $x \sin \phi + y \sin \theta - z = 0$.

Origin is common point on the above planes and if l, m, n be the direction-cosines of the line represented by them, then

$$\frac{l}{\sin \psi + \sin \theta \sin \phi} = \frac{m}{-\sin^2 \phi + 1} = \frac{n}{\sin \theta + \sin \phi \sin \psi}$$

If $\theta + \phi + \psi = 90$, then

$$\sin \psi = \sin \{(\pi/2) - (\theta + \phi)\} = \cos (\theta + \phi)$$

or

$$\sin \psi = \cos \theta \cos \phi - \sin \theta \sin \phi.$$

Similarly $\sin \theta = \cos \phi \cos \psi - \sin \phi \sin \psi$.

$$\therefore \frac{l}{\cos \theta \cos \phi} = \frac{m}{\cos^2 \phi} = \frac{n}{\cos \phi \cos \psi}$$

or

$$\frac{l}{\cos \theta} = \frac{m}{\cos \phi} = \frac{n}{\cos \psi}$$

Hence the equations of the line of intersection of (1) and (2)

$$\text{are } \frac{x}{\cos \theta} = \frac{y}{\cos \phi} = \frac{z}{\cos \psi}.$$

This line will lie in the third plane if the point $(0, 0, 0)$ on it lies on the plane which is true and if it is perpendicular to the normal to the plane whose d.c.'s are $\sin \psi, -1, \sin \theta$

or if $\cos \theta \sin \psi - \cos \phi + \sin \theta \cos \psi = 0$

or $\sin (\theta + \psi) - \cos \phi = 0$, put $\theta + \phi + \psi = \pi/2$,

or $\sin (\pi/2 - \phi) - \cos \phi = 0$ or $\cos \phi - \cos \phi = 0$.

which is true.

Ex. 11. The plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ meets the axes in A, B and C .

Prove that the planes through the axes and the internal bisectors of the angles of the triangle ABC pass through the line

$$\frac{x}{a\sqrt{(b^2+c^2)}} = \frac{y}{b\sqrt{(c^2+a^2)}} = \frac{z}{c\sqrt{(a^2+b^2)}} \quad [\text{Rohailkhand 83}]$$

$A(a, 0, 0); B(0, b, 0), C(0, 0, c).$

Direction-cosines of AB are $\frac{a}{\sqrt{(a^2+b^2)}}, \frac{-b}{\sqrt{(a^2+b^2)}}, 0$ and

those of AC are $\frac{a}{\sqrt{(a^2+c^2)}}, 0, \frac{-c}{\sqrt{(a^2+c^2)}}.$ Now we know that

if $l_1, m_1, n_1; l_2, m_2, n_2$ are the direction cosines of two lines, then the direction-cosines of internal bisectors of the lines are

$\frac{l_1+l_2}{2}, \frac{m_1+m_2}{2}, \frac{n_1+n_2}{2}.$ Q. 13 (a) P. 30. Hence the direction

cosines of the internal bisector of AB and AC , are

$$\frac{a}{2} \left\{ \frac{1}{\sqrt{(a^2+b^2)}} + \frac{1}{\sqrt{(a^2+c^2)}} \right\}, \frac{-b}{2\sqrt{(a^2+b^2)}}, \frac{-c}{2\sqrt{(a^2+c^2)}}.$$

say l, m, n .

Any plane through x -axis, i.e. $y=0$ and $z=0$ is $y+\lambda z=0$. If it contains the internal bisector whose d.c.'s are l, m, n the normal to the plane whose D.R.'s are $0, 1, \lambda$ will be perpendicular to it.

$$\therefore 0.l + 1.m + \lambda n = 0; \therefore \lambda = -m/n.$$

Hence the required plane is $y - \frac{m}{n} z = 0$ or $\frac{y}{m} = \frac{z}{n}$.

Putting for m and n the equation of the required plane is

$$\frac{-c}{2\sqrt{(a^2+c^2)}} y + \frac{b}{2\sqrt{(a^2+b^2)}} z = 0 \text{ or } \frac{y}{b\sqrt{(c^2+a^2)}} = \frac{z}{c\sqrt{(a^2+b^2)}}. \quad \dots(3)$$

Similarly we can write down the equation of the other planes

as

$$\frac{z}{c\sqrt{(a^2+b^2)}} = \frac{x}{a\sqrt{(b^2+c^2)}} \quad \dots(4)$$

$$\frac{x}{a\sqrt{(b^2+c^2)}} = \frac{y}{b\sqrt{(c^2+a^2)}} \quad \dots(5)$$

The planes (3), (4) and (5) clearly intersect on the line

$$\frac{x}{a\sqrt{(b^2+c^2)}} = \frac{y}{b\sqrt{(c^2+a^2)}} = \frac{z}{c\sqrt{(a^2+b^2)}}.$$

Ex. 12. The direction-cosines of OA, OB, OC are l_r, m_r, n_r , $r=1, 2, 3$ respectively and OA', OB', OC' bisect the angles BOC, COA, AOB . Show that the planes AOA', BOB', COC' pass through the line

$$\frac{x}{l_1+l_2+l_3} = \frac{y}{m_1+m_2+m_3} = \frac{z}{n_1+n_2+n_3}$$

The d.c.'s of bisector OA' of line OB and OC are

$$\frac{l_2+l_3}{2}, \frac{m_2+m_3}{2}, \frac{n_2+n_3}{2} \text{ so that its equations are}$$

$$\frac{x}{l_2+l_3} = \frac{y}{m_2+m_3} = \frac{z}{n_2+n_3} \quad \dots(1)$$

$$\text{Equations of } OA \text{ are } \frac{x}{l_1} = \frac{y}{m_1} = \frac{z}{n_1} \quad \dots(2)$$

Equation of a plane AOA' containing (1) and (2) is

$$\begin{vmatrix} x & y & z \\ l_1 & m_1 & n_1 \\ l_2+l_3 & m_2+m_3 & n_2+n_3 \end{vmatrix} = 0 \text{ [Apply } R_3+R_2]$$

$$\text{or } \begin{vmatrix} x & y & z \\ l_1 & m_1 & n_1 \\ l_1+l_2+l_3 & m_1+m_2+m_3 & n_1+n_2+n_3 \end{vmatrix} = 0 \quad \dots(3)$$

Similarly we can write planes BOB' and COC' .

The plane AOA' clearly passes through the line

$$\frac{x}{l_1+l_2+l_3} = \frac{y}{m_1+m_2+m_3} = \frac{z}{n_1+n_2+n_3} \quad \dots(4)$$

because if we substitute the general co-ordinates of any point on (4) in (3) it will be satisfied, as first and third rows in (3) will become identical and determinant vanishes. Similarly BOB' and COC' will also pass through the line (4).

Ex. 13. The plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ meets axes of co-ordinates (rectangular) in A, B and C . Find the equations of BC . Prove that the planes through the axes and perpendicular to BC, CA, AB pass through the line $ax=by=cz$.

Prove that the co-ordinates of the ortho-centre of the triangle ABC are

$$\left(\frac{a^{-1}}{a^{-2}+b^{-2}+c^{-2}}, \frac{b^{-1}}{a^{-2}+b^{-2}+c^{-2}}, \frac{c^{-1}}{a^{-2}+b^{-2}+c^{-2}} \right)$$

The points A, B, C are

$$(a, 0, 0), (0, b, 0), (0, 0, c)$$

$$\text{Equation to } BC \text{ is } \frac{x-0}{0} = \frac{y-0}{b} = \frac{z-c}{-c}$$

Plane through OX , i.e. $y=0$ and $z=0$ is $y+\lambda z=0$.

D.R.'s of normal are $0, 1, \lambda$. It is perpendicular to BC if normal is parallel to BC and hence the direction-ratios should be proportional.

$$\therefore \frac{0}{0} = \frac{b}{1} = \frac{-c}{\lambda} \quad \text{or } \lambda = \frac{-c}{b}$$

Putting for λ the equations of plane through OX and perpendicular to BC is $by-cz=0$.

Similarly the equations of the other planes are $ax=by, cz=ax$ and these planes clearly intersect in the line.

$$ax=by=cz=r \text{ say.}$$

Also the orthocentre lies on the three planes and it also lies on the plane of $\triangle ABC$, i.e. $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

In other words, the orthocentre will lie on the line of intersection of the planes, i.e. $ax=by=cz=r$ say and the plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$. Thus it is the intersection of the line and the plane.

Any point on the line is $\left(\frac{r}{a}, \frac{r}{b}, \frac{r}{c}\right)$. If it lies on the plane,

$$\text{then } \left(\frac{r}{a}+\frac{r}{b}+\frac{r}{c}\right)=1; \therefore r=\frac{1}{a^{-2}+b^{-2}+c^{-2}}.$$

Putting the value of r , the co-ordinates of the ortho-centre are

$$\left(\frac{a^{-1}}{a^{-2}+b^{-2}+c^{-2}}, \frac{b^{-1}}{a^{-2}+b^{-2}+c^{-2}}, \frac{c^{-1}}{a^{-2}+b^{-2}+c^{-2}}\right)$$